

PAPER_1334 — JWST High-z Massive Galaxy Excess

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "jwst_high_z_excess"})`

Master Expression

SF efficiency boost = $K_{\text{MEX}} \times \Phi_{\text{res}} = 1.75$

Match: covers JWST z=14 detections

Interpretation

$F_{\text{U_Bi_i}}$ buoyancy amplification at high z produces early massive galaxies without Λ CDM modification.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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